CTRLMOL: CONTROLLABLE MOLECULAR GENERATION FOR DRUG DESIGN VIA FLEXIBLE CONDITION-ADAPTIVE DIFFUSION

Anonymous authors
Paper under double-blind review

ABSTRACT

Conditional molecular generation is critical for drug design, where generated molecules need to satisfy varying design requirements. However, existing molecular generation methods lack flexibility in handling diverse reference conditions and controllability in adjusting generation toward desired molecular profiles, limiting their ability to meet some customized requirements. To address these limitations, we propose CtrlMol, a novel controllable molecular generation framework via flexible condition-adaptive diffusion model. CtrlMol achieves flexibility through Arbitrary-Count Condition Encoding that accommodates zero or more retrieved molecular and pocket conditions, followed by Self-Adaptive Modulation aggregating them into the denoising process in a unified manner. CtrlMol also achieves controllability through Independent Dual-Axis Classifier-Free Guidance that separately adjusts the guidance strengths of molecular and pocket conditions during inference, allowing controllable generation trade-offs between precision and diversity. Experiments on the TarPass benchmark demonstrate that CtrlMol consistently outperforms competing methods, achieving 28.8% improvement in target specificity and superior molecular property control. Case studies on a cardiovascular target ADRB2 further validate that the flexibility and controllability jointly enable effective adjustment of the generation distribution.

1 INTRODUCTION

Molecular generation is a central task in drug design, aiming to discover candidate compounds with desired properties and biological activities (Sun et al., 2025). In recent years, deep learning-based conditional molecular generation methods have significantly accelerated this process by encoding specific design requirements into conditional signals that guide generation (Sousa et al., 2021; Du et al., 2024). In practical drug discovery, two types of prior knowledge as conditions are particularly valuable. Known active ligands provide desired molecular property profiles and support ligand-based drug design (López-López et al., 2026), while known binding pockets from protein-ligand complexes provide binding interaction information and support structure-based drug design (Pala & Clark, 2024). Integrating both conditions is essential for generating molecules with desired properties and target specificity.

However, in real-world scenarios, the types and counts of available conditions often vary depending on the design requirements or data availability. Consequently, an ideal molecular generation system should flexibly adapt to different condition configurations and provide control over the generation process during inference. In practice, when faced with a new drug design task, researchers generally begin by retrieving relevant known active ligands and target binding pocket information from databases, using them as reference conditions to guide the generation of novel molecules. However, the counts of retrieved reference ligands and target pockets are uncertain or even absent, and the two types of conditions are often unrelated or unpaired. Existing conditional molecular generation methods, whether accepting ligands (Li et al., 2026), pockets (Guan et al., 2023), or paired ligand-pocket complexes (Cremer et al., 2026) as conditions, are typically designed to accept fixed types and fixed counts of condition inputs, making it difficult to directly adapt to such variability in the types and counts of condition configurations in real-world retrieval scenarios.

Reviewers: please read the Reviewer Guidelines (iclr.cc/Conferences/2027/ReviewerGuidelines) and the AI Policy for Reviewers (iclr.cc/Conferences/2027/AIPolicyForReviewers). If you used AI to expand, edit, or polish your review, please provide the input text to the LLM. Better still, consider skipping the LLM and submitting your original text: we, and the authors, are much more interested in your unedited thoughts than in what an LLM has to say. AI-assisted or not, you are putting your name and reputation behind your review: LLM-generated falsehoods, hallucinations or misrepresentations are subject to disciplinary action, which may include desk-rejecting all papers you have authored.

On the other hand, without controllability over condition influence, overly strict adherence to reference conditions would eliminate the exploratory power of generative models. An ideal generation system should be able to control how closely generated molecules match the conditions to balance fidelity and novelty. Existing methods (Ünlü et al., 2025; Xie et al., 2025; Zhang et al., 2024) are commonly trained with fixed condition configurations and cannot adjust guidance strengths at inference time, thus lacking the ability to control desired generation behavior. This mismatch between fixed input requirements and variable practical needs limits the utility of existing methods in real-world drug design.

To address these challenges, we propose CtrlMol, a controllable molecular generation framework via flexible condition-adaptive diffusion model. CtrlMol supports zero or more reference molecular conditions and/or target pocket conditions as input via an Arbitrary-Count Condition Encoding mechanism, and natively adapts to different condition configurations within a unified framework to meet customized design requirements. Through a Self-Adaptive Condition Modulation mechanism, CtrlMol adaptively aggregates heterogeneous condition information of different counts and types, and injects them into the diffusion process in a unified manner. At inference time, CtrlMol employs an Independent Dual-Axis Classifier-Free Guidance mechanism that allows users to separately adjust the guidance strengths of molecular and pocket conditions, flexibly controlling the generation direction toward desired design requirements. Extensive experiments on the TarPass benchmark demonstrate that CtrlMol consistently outperforms existing methods across basic performance, molecular properties, and achieves a 28.8% improvement over the second-best baseline in target specificity. Further case studies on a cardiovascular target ADRB2 and systematic ablation studies confirm that the flexibility in handling arbitrary-count conditions and the controllability in adjusting dual-axis guidance scales jointly enable effective tuning of generation distribution toward desired molecular profiles.

Our main contributions can be summarized as follows:

- We propose CtrlMol, a controllable molecular generation framework via flexible condition-adaptive diffusion model, shifting drug design from developer-centric technical implementations to user-centric practical design requirements.
- We introduce Arbitrary-Count Condition Encoding (ACCE) and Self-Adaptive Condition Modulation (SACM) mechanisms that jointly enable flexible handling of variable or absent conditions and their adaptive aggregation for unified modulation throughout the denoising process.
- We develop Independent Dual-Axis Classifier-Free Guidance (IDA-CFG) to separately control the guidance strengths of molecular and pocket conditions during inference, achieving flexible adjustment of the generation fidelity-diversity trade-off.

2 Related Work

2.1 Diffusion Models for Molecule Generation

Diffusion models (Ho et al., 2020) have been widely adopted for molecular generation tasks (Hoogeboom et al., 2022) in recent years following their success in image generation, demonstrating superior performance over VAEs and GANs in generation quality and sample diversity (Sun et al., 2025). Early molecular generation primarily focused on generating SMILES strings or 2D molecular graphs, while recent research has shifted toward directly generating 3D molecular conformations comprising atomic coordinates, atom types, and bond types (Sun et al., 2025). To ensure that generated molecular coordinates are equivariant to translation and rotation, these methods typically employ SE(3)-equivariant network architectures (Liao et al., 2026; Aykent & Xia, 2025) and combine hybrid diffusion processes that separately model continuous atomic coordinates and discrete atom types and bond types (Peng et al., 2026). While unconditional diffusion models generate diverse molecules, they lack specificity for practical drug design, which requires conditional guidance to generate molecules satisfying particular property requirements or target binding constraints. Existing conditional molecular diffusion methods typically inject condition information into the denoising network during training (Huang et al., 2024), or achieve conditional control during inference through gradient guidance based on energy functions (Han et al., 2024) or predictive models (Chen et al., 2025).

2.2 Conditional Molecular Generation

The key challenge in drug design lies in understanding drug-target interactions(Zhao et al., 2025b), which result in two major paradigms of conditional generation: ligand-based drug design (LBDD) and structure-based drug design (SBDD). LBDD methods use information from reference ligands as conditions to guide the generation of similar novel molecules, including molecular physicochemical properties (e.g., QED, SA) (Hu et al., 2025), molecular structures or fragments (e.g., scaffolds, pharmacophores) (Peng et al., 2025), molecular-level features or fingerprints (e.g., Morgan fingerprints, graph embeddings) (Joshi & Guven, 2025). These methods typically accept fixed types and fixed counts of conditions as input. SBDD methods use information from target proteins as conditions to guide the generation of novel molecules for binding to specific targets, including target sequences (e.g., protein language model embeddings) (Su et al., 2026) and target pocket structures (e.g., 2D surface embeddings, 3D geometry embeddings) (Isert et al., 2023). These methods typically allow only a single target condition input at a time. Very few works attempt to combine LBDD and SBDD by leveraging both reference ligand and target pocket information, but these methods still require fixed types and fix counts of conditions, and even require ligand-pocket interaction information as additional conditions (Cremer et al., 2026). However, such requirements are impractical in generation scenarios, as interaction information cannot be predetermined before the molecule is generated.

In recent years, Retrieval-Augmented Generation (RAG) has achieved remarkable success in natural language processing (NLP) by dynamically retrieving relevant information from knowledge bases, flexibly handling arbitrary types and counts of retrieval conditions as prompt (Guo et al., 2025). However, RAG has rarely been introduced into non-NLP-based molecular generation models (Chan et al., 2026), as these models struggle to handle variable counts of retrieval conditions, despite the fact that retrieving reference ligands and target structures is the prerequisite step in practical drug design.

2.3 Conditioning Mechanisms in Diffusion Models

Conditioning mechanisms determine how condition information is incorporated into the denoising network of diffusion models, directly affecting the responsiveness to conditions. Early diffusion models simply concatenate condition embeddings with hidden representations and inject them into the denoising network (Saharia et al., 2023), which requires fixed counts of conditions and lacks active control over condition embeddings. Cross-attention mechanism treats condition embeddings as key-value pairs and hidden representations as queries for interaction (Rombach et al., 2022), enabling handling variable counts of conditions, but the computational complexity is inherently large and scales linearly with the count of conditions. Feature-wise Linear Modulation (FiLM) linearly projects condition embeddings into scale and shift parameters for affine transformation of hidden representations (Perez et al., 2018), achieving direct control over conditions compared to the concatenation mechanism and lower computational complexity compared to the cross-attention mechanism. Although original FiLM still requires fixed counts of conditions, a series of adaptive variants (e.g., AdaLN (Peebles & Xie, 2023)) have been widely adopted in diffusion models. The controllability, low complexity, and extensibility of FiLM provide potential for adaptation to handle variable counts of conditions.

2.4 Controllable Generation via Guidance

Guidance mechanisms enable diffusion models to control the influence of conditions at inference time, without requiring model re-training. Classifier guidance uses gradients from external classifiers to direct sampling toward desired conditions at inference (Dhariwal & Nichol, 2021), but requires training a separate classifier for each condition type. Classifier-free guidance (CFG) eliminates the need for external classifiers by jointly training the model on both conditional and unconditional objectives with a dropout rate (Ho & Salimans, 2022), but requires a separate guidance scale to adjust each condition type. The theory of one-dimensional CFG with one condition type is relatively well-understood, and a few studies have applied it to molecular generation (Worakul et al., 2026; Wang et al., 2025; Hu et al., 2025). However, two- or higher-dimensional CFG with multiple condition types suffers from a combinatorial explosion, leaving its theory underdeveloped (Schnell et al., 2025) or even contested (Pavasovic et al., 2025; Ventura et al., 2026), and its applications remain extremely rare. CFG and its variants have become the mainstream guidance mechanism for condi-

tional diffusion models owing to their flexible adjustment of condition strength, while independent guidance on multiple condition types is a genuine requirement in real-world applications.

3 METHOD

3.1 PROBLEM FORMULATION

We study the task of molecular generation guided by two types of conditions: a set of reference molecules $\mathcal{M} = \{m_1, \ldots, m_K\}(K \geq 0)$ and a set of reference target pockets $\mathcal{P} = \{p_1, \ldots, p_L\}(L \geq 0)$. We represent a molecule as a triplet (X, A, B), where X denotes atomic coordinates, A denotes atom types, and B denotes bond types. We formulate the task by learning a conditional distribution $p_\theta(X, A, B \mid \mathcal{M}, \mathcal{P})$ and sampling chemically valid, geometrically reasonable molecules from it. Both $\mathcal{M}$ and $\mathcal{P}$ are mutually independent and can be arbitrary or absent. Our model natively handles all condition configurations within a unified framework.

3.2 OVERVIEW

We present CtrlMol, an end-to-end framework for controllable molecular generation via flexible condition-adaptive diffusion. As illustrated in Figure 1, the pipeline takes a SMILES string as input, encodes an initial 3D conformer, diffuses and denoises the 3D molecular structure under independent dual-axis guidance from arbitrary-count molecular and pocket conditions, finally decodes the resulting structure back into a SMILES string.

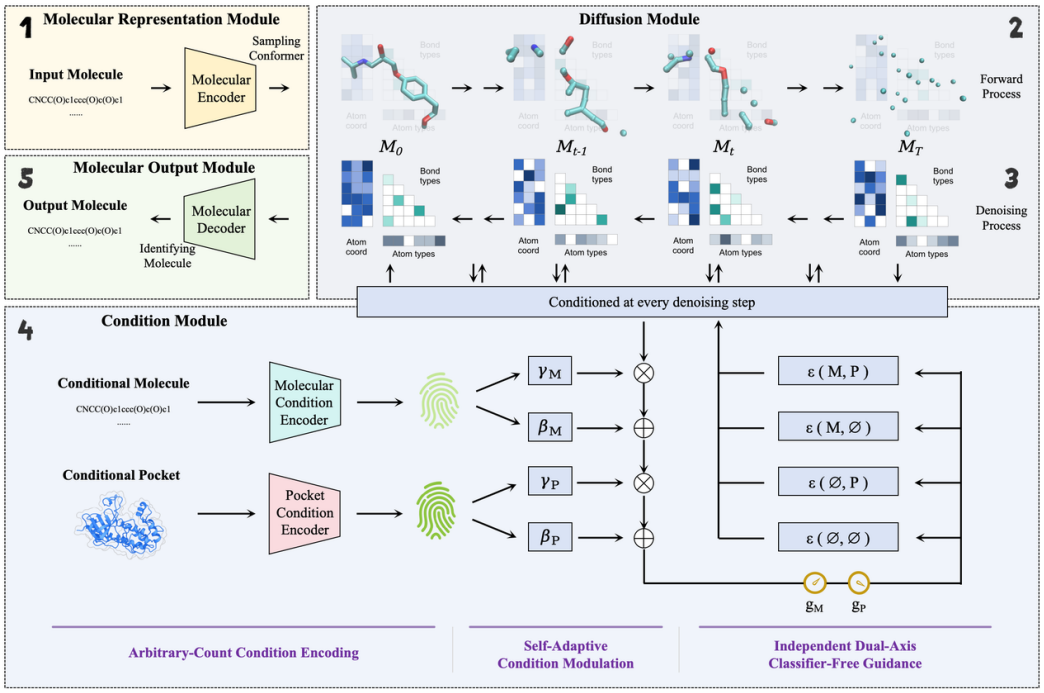


Figure 1: **Overview of CtrlMol framework.** Training follows steps 1-2-3-4, inference follows steps 3-4-5.

Molecular Representation Module. During training, each molecule in the dataset is provided as a SMILES string and converted by a molecular encoder into a 3D conformer representation (X, A, B). Details are provided in Section 3.3.

Diffusion Module. A hybrid diffusion process is employed: continuous diffusion for atomic coordinates X, and discrete diffusion for atom types A and bond types B. The forward process gradually corrupts the real molecule (X, A, B) into noise; the reverse process reconstructs a structured

molecule $(\hat{X}, \hat{A}, \hat{B})$ from noise, guided at every step by the condition system. Details are provided in Appendix A.

Condition Module. Conditions serve as the key mechanism for guiding molecular generation toward specific design requirements. Given a drug design task, relevant reference molecules and target pockets are retrieved from databases to serve as two set of conditions. Through Arbitrary-Count Condition Encoding, a set of retrieved reference molecular SMILES are encoded by a molecular condition encoder into a set of molecular fingerprints; a set of retrieved target pocket structures are encoded by a pocket condition encoder into a set of pocket fingerprints. The two sets of fingerprints are respectively aggregated by Self-Adaptive Condition Modulation into the denoising network in a unified manner at every step. Through Independent Dual-Axis Classifier-Free Guidance, the guidance strengths of molecular and pocket conditions can be independently controlled to adjust generation toward design requirements. Details are provided in Section 3.4.

Molecular Output Module. The denoised $(\hat{X}, \hat{A}, \hat{B})$ is converted by a molecular decoder into a SMILES string. This process checks chemical validity and rationality, and the generation is deemed unsuccessful if the denoised result violates chemical rules.

3.3 Molecular Representation and Conformer Generation

Molecular Representation. A molecule is represented by the triplet (X, A, B).

$X \in \mathbb{R}^{N\times 3}$ denotes the 3D coordinates of N heavy atoms.

$A \in \{\varnothing, \text{C, O, N, P, S, X}\}^N$ denotes atom types, where $\varnothing$ represents a null atom (no atom at that position), C/O/N/P/S represent common elements, X unifies halogens F/Cl/Br/I, and hydrogens are implicitly encoded.

$B \in \{\varnothing, -, =, \#, :\}^{N(N-1)/2}$ denotes bond type matrix: null bond ($\varnothing$), single bond ($-$), double bond ($=$), triple bond ($\#$), and aromatic bond ($:$).

Flexible-size Molecular Generation. Existing molecular generation methods typically require specifying the number of atoms before inference (Ünlü et al., 2025; Peng et al., 2026; Zhang et al., 2025), which is impractical since molecular size is unknown beforehand. By introducing null atom type and null bond type, our model eliminates this impractical requirement. It autonomously determines molecular size within a fixed-dimensional tensor, dynamically deciding during denoising which positions correspond to real atoms and which remain vacant. We set $N = 50$ in this work, enabling generation of molecules with up to 50 heavy atoms.

Conformer Generation. The SMILES is first parsed into a molecular graph (A, B) via RDKit (Landrum et al.). Initial 3D coordinates are then randomly generated for the molecule, followed by energy minimization with the MMFF94s force field (Halgren, 1999) to obtain optimized 3D coordinates X. Through this process, a SMILES string is encoded into a randomly initialized 3D conformer.

On-the-Fly Conformer Generation. Common training datasets (Tingle et al., 2023; Knox et al., 2024) maintain a one-to-one mapping between molecular formula and 3D conformer, assuming each molecule has a unique ground-state structure. In reality, however, molecules adopt different conformations depending on their binding environment (Zhao et al., 2025a). The same molecule can exhibit multiple conformations when binding to different targets, or even to the same target. To address this, we generate conformers on-the-fly during training rather than drawing from a fixed dataset. Each time a molecule is encoded, a distinct conformer is randomly produced, implicitly augmenting the training data with conformational diversity. This approach encourages the model to learn the chemical essence of atomic connectivity rather than memorize specific conformers, yielding stronger generalization to unseen conformations at inference.

3.4 Flexible Condition Encoding and Controllable Guidance

Conditions are integrated into the denoising network at every time step to guide the generation process.

3.4.1 ARBITRARY-COUNT CONDITION ENCODING

Given a specific drug design task, we retrieve relevant reference molecules from any of chemical databases (e.g., ChEMBL (Zdrazil et al., 2024), PubChem (Kim et al., 2025), BindingDB(Liu et al., 2025)) and target binding pockets from any of structural databases (e.g., UniProt (Consortium, 2025), PDB (wwPDB consortium, 2019), AlphaFold DB (Varadi et al., 2022)). The retrieval yields a reference molecule set $\mathcal{M} = \{m_1, \ldots, m_K\}$ and a target pocket set $\mathcal{P} = \{p_1, \ldots, p_L\}$, where both K and L are arbitrary non-negative integers determined by retrieval results and are mutually independent.

Each molecule m_i (provided as a SMILES string) is independently encoded by a molecular condition encoder $f_{\mathcal{M}}$ into a fingerprint $\phi_i = f_{\mathcal{M}}(m_i)$, yielding a molecular fingerprint set $\Phi_{\mathcal{M}} = \{\phi_1, \ldots, \phi_K\}$. The molecular condition encoder can be a trainable model or any existing molecular encoder (e.g., ECFP (Rogers & Hahn, 2010), ChemNet (Mayr et al., 2018), Uni-Mol (Ji et al., 2024)). Likewise, each pocket p_j (represented as a 3D structure) is independently encoded by a pocket condition encoder $f_{\mathcal{P}}$ into a fingerprint $\psi_j = f_{\mathcal{P}}(p_j)$, yielding a pocket fingerprint set $\Psi_{\mathcal{P}} = \{\psi_1, \ldots, \psi_L\}$. The pocket condition encoder can also be a trainable model or any existing pocket encoder (e.g., MaSIF (Gainza et al., 2020), DrugCLIP (Jia et al., 2026)). Empty condition set ($K = 0$ or $L = 0$) is regarded as zero padding vector.

3.4.2 SELF-ADAPTIVE CONDITION MODULATION

Each set of arbitrary-count fingerprints must be adaptively integrated into the denoising network in a unified manner. The standard FiLM mechanism (Perez et al., 2018) maps a condition vector y through an MLP to channel-wise scale and shift parameters $(\gamma, \beta) = \mathrm{split}\big(W_2\,\mathrm{SiLU}(W_1 y)\big)$, which affinely modulate the denoising network's hidden representation as $z \leftarrow \gamma \odot z + \beta$. By reparameterizing the modulation as $z \leftarrow (1+\gamma) \odot z + \beta$, the mechanism naturally degenerates to identity $z \leftarrow z$ when the condition is absent ($y = 0$), natively supporting zero-condition cases. Furthermore, given an arbitrary-count condition set $y = \{y_1, ..., y_N\}$, each condition y_i can independently generate its modulation parameters through the MLP, which are then aggregated into unified parameters (γ, β) and applied to the hidden representation z.

$$(\gamma_i, \beta_i) = \mathrm{split}\big(W_2, \mathrm{SiLU}(W_1 y_i)\big), \quad i = 1, ..., N$$
$$(\gamma, \beta) = \big(\mathrm{Agg}(\gamma_1, ..., \gamma_N), \mathrm{Agg}(\beta_1, ..., \beta_N)\big)$$

For molecular conditions $\mathcal{M}$ and pocket conditions $\mathcal{P}$, we independently apply the aggregation mechanism to each type, yielding two sets of modulation parameters $(\gamma_{\mathcal{M}}, \beta_{\mathcal{M}})$ and $(\gamma_{\mathcal{P}}, \beta_{\mathcal{P}})$. We implement aggregation as mean pooling. The final modulation averages the two branches

$$z \leftarrow \frac{1}{2}\big[(1 + \gamma_{\mathcal{M}}) \odot z + \beta_{\mathcal{M}} + (1 + \gamma_{\mathcal{P}}) \odot z + \beta_{\mathcal{P}}\big]$$

When either condition type is absent, the corresponding parameters become zero, reducing to single-type modulation $z \leftarrow (1 + \gamma) \odot z + \beta$. When both are absent, it naturally degenerates to identity $z \leftarrow z$. This mechanism self-adaptively modulates the hidden representation given arbitrary-count or absent conditions at every denoising step.

3.4.3 INDEPENDENT DUAL-AXIS CLASSIFIER-FREE GUIDANCE

Classifier-free guidance (CFG) (Ho & Salimans, 2022; Rombach et al., 2022) trains the model by randomly replacing the condition c by $\varnothing$ with probability p_{drop}, enabling the model to learn both conditional predicted noise $\epsilon_\theta(x_t, t, c)$ and unconditional predicted noise $\epsilon_\theta(x_t, t, \varnothing)$. At inference, CFG extrapolates from the unconditional toward the conditional predicted noise with guidance scale w, yielding guided noise estimate

$$\hat{\epsilon} = \epsilon_\theta(x_t, t, \varnothing) + w \cdot (\epsilon_\theta(x_t, t, c) - \epsilon_\theta(x_t, t, \varnothing)) .$$

However, the condition c accommodates only a single condition type.

To independently control the guidance strengths of molecular conditions $\mathcal{M}$ and pocket conditions $\mathcal{P}$, we extend the standard CFG to a dual-axis formulation. During training, $\mathcal{M}$ and $\mathcal{P}$ are independently dropped with probabilities $p_{\mathcal{M}}$ and $p_{\mathcal{P}}$, enabling the model to learn all four condition combinations: $(\mathcal{M}, \mathcal{P})$, $(\mathcal{M}, \varnothing)$, $(\varnothing, \mathcal{P})$, and $(\varnothing, \varnothing)$. At inference, two independent guidance scales $g_{\mathcal{M}}$

and $g_{\mathcal{P}}$ control the contribution of each condition axis, yielding the guided estimate

$$\hat{\epsilon} = (1-\alpha-\beta_1-\beta_2)\epsilon_\theta(x_t, t, \varnothing, \varnothing) + \alpha\epsilon_\theta(x_t, t, \mathcal{M}, \mathcal{P}) + \beta_1\epsilon_\theta(x_t, t, \mathcal{M}, \varnothing) + \beta_2\epsilon_\theta(x_t, t, \varnothing, \mathcal{P}),$$

where $\alpha = \min(g_{\mathcal{M}}, g_{\mathcal{P}})$, $\beta_1 = g_{\mathcal{M}} - \alpha$, and $\beta_2 = g_{\mathcal{P}} - \alpha$. This dual-axis CFG enables independent control of molecular and pocket guidance strengths, allowing flexible adjustment of generation toward specific design requirements.

4 Experiments

4.1 Experimental Setup

Datasets. We trained CtrlMol on a combination of ChEMBL and BindingDB. ChEMBL provides broad chemical diversity (2.8 M molecules), while BindingDB contributes target-ligand binding annotations (2.7 M pairs), jointly covering both molecular properties and binding knowledge. The same datasets serve as the retrieval source for molecular and pocket conditions during both training and inference, and can be freely replaced by other databases. We evaluated CtrlMol on TarPass (Qin et al., 2026), an external benchmark for fair evaluation of target-aware de novo molecular generation models, comprising 18 well-studied targets and approximately 1,000 experimentally validated active compounds per target. Details are provided in Appendix B.1.

Evaluation Metrics. Following TarPass, we evaluated CtrlMol across four complementary aspects: basic performance (validity, diversity), molecular property (Wasserstein distance over 15 molecular descriptors), binding affinity (Vina docking score, ligand efficiency), and target specificity (ECFP4 Tanimoto similarity, Fréchet ChemNet Distance (FCD) distance and shift). Distance means the distance between generated distribution and reference distribution. Shift means the normalized distance of generated distribution from both reference and random distributions. Details are provided in Appendix B.2.

Baselines. Following TarPass, we compared against 15 representative molecular generation models spanning three paradigms: (i) non-3D models (DeepBlock, DRAGONFLY, SimpleSBDD, and TamGen), which generate 2D molecular graphs or strings without explicit structural geometry; (ii) 3D in situ models (DiffSBDD, DrugFlow, IPDiff, Lingo3DMol, MolCraft, PocketFlow, SurfGen, and TargetDiff), which operate directly on target pocket structures; and (iii) optimization-based methods (DrugFlow-PA, MolPilot, and REINVENT), which further optimize generated molecules toward desired properties to guide sampling. Details are provided in Appendix B.3.

4.2 Main Results

We present the main results on TarPass benchmark across 18 diverse targets in Table 1. CtrlMol achieves perfect validity and diversity while demonstrating the best performance in molecular property control and target specificity among all baselines. CtrlMol achieves perfect validity (1.0) and diversity (1.0), meeting the basic generation quality requirements. For target specificity measured by FCD and Tanimoto similarity to target-binding reference molecules (Appendix B.5), CtrlMol substantially outperforms all baselines with the lowest FCD distances (FCD Ref = 16.351, FCD Shift = 1.210), representing 28.8% improvement over the second-best DeepBlock, and competitive molecular similarity (Sim. Ref = 0.128), demonstrating superior ability to capture target-specific chemical space. For molecular property control measured by Wasserstein distance over 15 molecular descriptors (Appendix B.6), CtrlMol also achieves the best performance (Wass. Ref = 0.497, Wass. Shift = 1.760).

For binding affinity, CtrlMol outperforms all four non-3D methods (DRAGONFLY et al.), achieves comparable performance to eight 3D in situ methods (DiffSBDD et al.), but lags behind all three optimization-based methods (DrugFlow-PA et al.). This gap highlights the effectiveness of explicit optimization techniques in improving binding affinity, suggesting a promising direction for future enhancement of CtrlMol. 3D in situ methods generate molecules directly within the fixed target pocket structures provided by the benchmark, naturally achieving stronger binding affinities towards the same pockets but limiting their ability to cover the broader target-specific chemical space required for dynamic binding in real-world scenarios. In contrast, CtrlMol decouples pocket and molecular conditions as independent guidance with arbitrary-count retrieval, naturally favoring

broader target-specific chemical space coverage over rigid target docking. On this basis, CtrlMol can further leverage its flexible condition and controllable guidance mechanism to adjust the trade-off between coverage and binding.

Table 1: **Main results on TarPass benchmark.** CtrlMol is compared against 15 baseline models across basic performance, molecular property, binding affinity, and target specificity. The reported values are averaged across 18 targets (Appendix B.5).

	Basic Performance		Molecular Property		Binding Affinity		Target Specificity		
Model	**Validity**	**Diversity (Uniq@1k)**	**Descriptor Distance (↓)**		**Vina Score (Med., ↓)**	**Ligand Efficiency (↑)**	**Molecular Distance in Chemical Space (↓)**		
			Wass. Ref	**Wass. Shift**			**Sim. Ref (↑)**	**FCD Ref**	**FCD Shift**
DRAGONFLY	1	1	<u>0.629</u>	2.166	-8.192	0.261	0.127	27.131	1.443
DeepBlock	1	1	0.884	2.685	-7.776	0.339	0.120	<u>22.982</u>	<u>1.313</u>
SimpleSBDD	0.966	0.987	0.856	2.322	-8.230	0.319	0.112	39.009	2.665
TamGen	∼1	0.968	1.346	4.243	-6.890	0.357	0.103	36.648	2.078
DiffSBDD	0.968	0.987	1.170	3.957	-8.032	0.326	0.094	31.824	2.093
DrugFlow	0.831	0.399	0.633	2.512	-9.776	0.335	**0.130**	32.178	2.442
IPDiff	∼1	0.940	1.906	6.896	-7.718	**0.421**	0.069	50.292	3.830
Lingo3DMol	∼1	0.906	0.822	3.408	-9.024	0.285	0.122	29.541	2.131
MolCraft	1	0.498	0.763	2.649	-9.987	0.350	0.112	38.369	2.880
PocketFlow	1	0.692	1.414	4.722	-7.795	<u>0.412</u>	0.097	36.967	2.469
SurfGen	1	0.948	1.122	4.126	-9.531	0.377	0.110	42.562	3.245
TargetDiff	∼1	0.968	1.233	4.719	-7.583	0.399	0.096	44.594	3.497
DrugFlow-PA	0.600	0.637	0.955	3.658	<u>-10.374</u>	0.404	0.128	42.897	3.371
MolPilot	1	1	0.796	3.156	**-10.483**	0.338	0.119	41.495	3.208
REINVENT4	1	1	1.392	5.592	-10.023	0.367	0.103	49.921	3.890
CtrlMol	**1**	**1**	**0.497**	**1.760**	-8.510	0.309	<u>0.128</u>	**16.351**	**1.210**
Random	—	—	0.695	—	-7.835	0.293	0.116	26.984	—

4.3 Case Study and Ablation Study

Case Study on ADRB2 Drug Design. We selected β2-adrenergic receptor (ADRB2), a well-studied GPCR target for cardiovascular diseases (Rosenbaum et al., 2007), as a case study to demonstrate CtrlMol's flexibility and controllability in drug design. Our objective is to generate novel molecules covering the target-specific chemical space of known ADRB2 ligands. This addresses a key need in early drug discovery: exploring broad chemical diversity before selecting candidates for affinity optimization.

For ADRB2 drug design, we retrieved a natural substrate epinephrine, 13 clinical antagonists, 25 clinical agonists, and hundreds of experimental active ligands for available molecular conditions, as well as dozens of target-ligand binding structures for available pocket conditions (Table 2). Using these conditions, we investigated how retrieval conditions and guidance scales affect generation distribution through two ablation studies. For each experiment, we generate 1,000 molecules and measure FCD between the generated distribution and ADRB2 reference ligand distribution. Lower FCD indicates better coverage of the target-specific chemical space.

Ablation Study on Retrieval Conditions. We ablated the retrieval condition quantities by fixing guidance scales at $g_{\mathcal{M}} = g_{\mathcal{P}} = 1$ and systematically varying molecular condition count K and pocket condition count L to analyze their impact on generation quality (Table 2). When both conditions are absent ($K = 0, L = 0$), the generation distribution stays far from the ADRB2 reference distribution with an FCD of 27.42, but closely resembles a random drug-like distribution, reaching an FCD of 9.96 that almost approaches the minimum FCD achievable for a 1,000-molecule set (Appendix B.7). As the molecular condition count K increases with $L = 0$, FCD decreases steadily and plateaus at $\sim$ 19 once K reaches $\sim$ 100, indicating saturation thereafter. Notably, one molecular condition ($K = 1$) causes overfocused generation distribution that is also far from the reference (FCD = 28.33), meanwhile far from random distributions (FCD = 25.02), distinguishing it from unconditional generation (Appendix B.7). As the pocket condition count L increases with $K = 0$, FCD decreases slowly to $\sim$ 25 once L reaches $\sim$ 20, with only limited gains over the unconditioned

baseline. Interestingly, one pocket condition ($L = 1$) works better than several. Retrieval returns diverse binding conformations of the eligible ADRB2 target, and pocket fingerprints are sensitive enough to distinguish them. Aggregating them mixes conflicting geometric signals, which weakens the guidance toward the target-specific chemical space. When both conditions are present, the results are overall an interpolation of the two single-condition outcomes with limited synergistic gains, where molecular conditions dominate when $K \geq 50$, and pocket conditions dominate otherwise. In summary, retrieving approximately 100 molecular conditions and 1 pocket condition achieves near-optimal performance.

Table 2: FCD between generated molecules and ADRB2 reference ligands with fixed guidance scales ($g_{\mathcal{M}} = g_{\mathcal{P}} = 1$), and varying molecular condition count K and pocket condition count L.

	$K=0$	$K=1$	$K=13$	$K=25$	$K=50$	$K=100$	$K=200$	$K=300$
$L=0$	27.42	28.33	25.68	21.27	19.41	**19.25**	19.55	19.57
$L=1$	**23.18**	28.57	25.07	21.24	20.48	19.63	19.92	20.63
$L=4$	26.75	26.31	25.00	23.50	21.53	21.52	20.93	21.08
$L=10$	25.48	29.22	24.46	23.87	21.44	21.04	20.67	21.41
$L=20$	24.86	27.87	24.90	23.86	22.00	21.60	21.84	21.02

Ablation Study on Guidance Scales. We ablated the guidance scales by fixing retrieval condition quantities at $K = 300, L = 1$ and systematically varying $g_{\mathcal{M}}$ and $g_{\mathcal{P}}$ to analyze their impact on generation quality (Table 3). When both guidance scales vanish ($g_{\mathcal{M}} = 0, g_{\mathcal{P}} = 0$), the model degenerates to unconditional generation, consistent with results in Table 2. Across all configurations, $g_{\mathcal{M}} = 2$ generally achieves optimal performance fixing $g_{\mathcal{P}}$, while $g_{\mathcal{P}} = 1$ does likewise fixing $g_{\mathcal{M}}$. The two guidance scales jointly achieve optimal performance at $g_{\mathcal{M}} = 2, g_{\mathcal{P}} = 1$ with FCD = 17.52, demonstrating clear synergistic gains from independent dual-axis control (Appendix B.7 and B.8). Conversely, excessively high guidance scales degrade performance.

Table 3: FCD between generated molecules and ADRB2 reference ligands with fixed retrieval condition quantities ($K = 300, L = 1$), and varying guidance scales $g_{\mathcal{M}}$ and $g_{\mathcal{P}}$.

	$g_{\mathcal{M}}=0$	$g_{\mathcal{M}}=1$	$g_{\mathcal{M}}=2$	$g_{\mathcal{M}}=3$
$g_{\mathcal{P}}=0$	27.42	19.57	19.19	20.34
$g_{\mathcal{P}}=1$	23.18	20.63	**17.52**	19.31
$g_{\mathcal{P}}=2$	24.48	21.48	18.19	19.35
$g_{\mathcal{P}}=3$	26.35	21.99	18.75	19.15

5 Conclusion

In this work, we proposed CtrlMol, a controllable molecular generation framework via flexible condition-adaptive diffusion model that addresses the limitations of existing methods in handling diverse condition configurations and adjusting generation behavior. Through Arbitrary-Count Condition Encoding, Self-Adaptive Condition Modulation, and Independent Dual-Axis Classifier-Free Guidance, CtrlMol achieves both flexibility in accepting zero or more molecular and pocket conditions, and controllability in separately adjusting their guidance strengths during inference. Extensive experiments on the TarPass benchmark demonstrate that CtrlMol consistently outperforms existing methods across molecular properties and target specificity, while case studies on target ADRB2 confirm that the flexibility and controllability jointly enable effective adjustment of generation distribution toward desired molecular profiles. CtrlMol provides a promising direction toward shifting molecular generation from developer-centric technical implementations to user-centric practical design requirements. Future work may explore additional condition types and multi-objective optimization for broader drug design scenarios.

AI USE STATEMENT

In this work, we used generative AI tools for writing refinement and language polishing. We have not used generative AI tools for other tasks with required and recommanded disclosure, and the rest of the required disclosure and recommanded tasks are not applicable to this work. We have reviewed all AI-assisted work. All AI-edited text was manually checked by the authors against our original drafts to ensure the intended meaning was preserved. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

ETHICS STATEMENT

The authors confirm that this work complies with the ICLR Code of Ethics and raises no ethical concerns.

REPRODUCIBILITY STATEMENT

All experimental and computational details are described in Experiments and Appendix. All datasets used are publicly available. We will release the code upon publication.

REFERENCES

Sarp Aykent and Tian Xia. Gotennet: Rethinking efficient 3d equivariant graph neural networks. In Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu (eds.), *International Conference on Learning Representations*, volume 2025, pp. 40622–40646, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/file/64d4ff4fff788cdffe236f9ce8b09400-Paper-Conference.pdf.

Jonathan B. Baell and Georgina A. Holloway. New substructure filters for removal of pan assay interference compounds (pains) from screening libraries and for their exclusion in bioassays. *Journal of Medicinal Chemistry*, 53(7):2719–2740, 04 2010. doi: 10.1021/jm901137j. URL https://doi.org/10.1021/jm901137j.

A. Patrícia Bento, Anne Hersey, Eloy Félix, Greg Landrum, Anna Gaulton, Francis Atkinson, Louisa J. Bellis, Marleen De Veij, and Andrew R. Leach. An open source chemical structure curation pipeline using rdkit. *Journal of Cheminformatics*, 12(1):51, Sep 2020. ISSN 1758-2946. doi: 10.1186/s13321-020-00456-1. URL https://doi.org/10.1186/s13321-020-00456-1.

G. Richard Bickerton, Gaia V. Paolini, Jérémy Besnard, Sorel Muresan, and Andrew L. Hopkins. Quantifying the chemical beauty of drugs. *Nature Chemistry*, 4(2):90–98, Feb 2012. ISSN 1755-4349. doi: 10.1038/nchem.1243. URL https://doi.org/10.1038/nchem.1243.

Joey Chan, Wonbin Kweon, Ashley Shin, Niharika Bhattacharjee, Pengcheng Jiang, Yue Guo, and Jiawei Han. Mole-rag: Molecular structure-enhanced retrieval-augmented generation for chemistry, 2026. URL https://arxiv.org/abs/2606.05693.

Lianghong Chen, Dongkyu Eugene Kim, Mike Domaratzki, and Pingzhao Hu. Uncertainty-aware multi-objective reinforcement learning-guided diffusion models for 3d de novo molecular design. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL https://openreview.net/forum?id=lu4cGylISh.

The UniProt Consortium. Uniprot: the universal protein knowledgebase in 2025. *Nucleic Acids Research*, 53(D1):D609–D617, 01 2025. ISSN 1362-4962. doi: 10.1093/nar/gkae1010. URL https://doi.org/10.1093/nar/gkae1010.

Julian Cremer, Ross Irwin, Alessandro Tibo, Jon Paul Janet, Simon Olsson, and Djork-Arné Clevert. FLOWR: flow matching for structure-aware de novo, interaction- and fragment-based ligand generation. *Nature Computational Science*, 6(6):565–574, June 2026. ISSN 2662-8457. doi: 10.1038/s43588-026-00998-8. URL https://doi.org/10.1038/s43588-026-00998-8.

Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan (eds.), *Advances in Neural Information Processing Systems*, volume 34, pp. 8780–8794. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/49ad23d1ec9fa4bd8d77d02681df5cfa-Paper.pdf.

Yuanqi Du, Arian R. Jamasb, Jeff Guo, Tianfan Fu, Charles Harris, Yingheng Wang, Chenru Duan, Pietro Liò, Philippe Schwaller, and Tom L. Blundell. Machine learning-aided generative molecular design. 6(6):589–604, 2024. ISSN 2522-5839. doi: 10.1038/s42256-024-00843-5. URL https://doi.org/10.1038/s42256-024-00843-5.

P. Gainza, F. Sverrisson, F. Monti, E. Rodolà, D. Boscaini, M. M. Bronstein, and B. E. Correia. Deciphering interaction fingerprints from protein molecular surfaces using geometric deep learning. *Nature Methods*, 17(2):184–192, February 2020. ISSN 1548-7105. doi: 10.1038/s41592-019-0666-6. URL https://doi.org/10.1038/s41592-019-0666-6.

Jiaqi Guan, Wesley Wei Qian, Xingang Peng, Yufeng Su, Jian Peng, and Jianzhu Ma. 3d equivariant diffusion for target-aware molecule generation and affinity prediction. In *International Conference on Learning Representations*, 2023.

Zirui Guo, Xubin Ren, Lingrui Xu, Jiahao Zhang, and Chao Huang. Rag-anything: All-in-one rag framework, 2025. URL https://arxiv.org/abs/2510.12323.

Thomas A. Halgren. Mmff vi. mmff94s option for energy minimization studies. *Journal of Computational Chemistry*, 20(7):720–729, 1999. doi: https://doi.org/10.1002/(SICI)1096-987X(199905)20:7⟨720::AID-JCC7⟩3.0.CO;2-X. URL https://onlinelibrary.wiley.com/doi/abs/10.1002/%28SICI%291096-987X%28199905%2920%3A7%3C720%3A%3AAID-JCC7%3E3.0.CO%3B2-X.

Xu Han, Caihua Shan, Yifei Shen, Can Xu, Han Yang, Xiang Li, and Dongsheng Li. Training-free multi-objective diffusion model for 3d molecule generation. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=X41c4uB4k0.

Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance, 2022. URL https://arxiv.org/abs/2207.12598.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin (eds.), *Advances in Neural Information Processing Systems*, volume 33, pp. 6840–6851. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/file/4c5bcfec8584af0d967f1ab10179ca4b-Paper.pdf.

Emiel Hoogeboom, Didrik Nielsen, Priyank Jaini, Patrick Forré, and Max Welling. Argmax flows and multinomial diffusion: Learning categorical distributions. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan (eds.), *Advances in Neural Information Processing Systems*, volume 34, pp. 12454–12465. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/67d96d458abdef21792e6d8e590244e7-Paper.pdf.

Emiel Hoogeboom, Víctor Garcia Satorras, Clément Vignac, and Max Welling. Equivariant diffusion for molecule generation in 3D. In Kamalika Chaudhuri, Stefanie Jegelka, Le Song, Csaba Szepesvari, Gang Niu, and Sivan Sabato (eds.), *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 8867–8887. PMLR, 17–23 Jul 2022. URL https://proceedings.mlr.press/v162/hoogeboom22a.html.

Qiaoyu Hu, Changzhi Sun, Huan He, Jiazheng Xu, Danlin Liu, Wenqing Zhang, Sumeng Shi, Kai Zhang, and Honglin Li. Target-aware 3d molecular generation based on guided equivariant diffusion. *Nature Communications*, 16(1):7928, Aug 2025. ISSN 2041-1723. doi: 10.1038/s41467-025-63245-0. URL https://doi.org/10.1038/s41467-025-63245-0.

Zhilin Huang, Ling Yang, Xiangxin Zhou, Chujun Qin, Yijie Yu, Xiawu Zheng, Zikun Zhou, Wentao Zhang, Yu Wang, and Wenming Yang. Interaction-based retrieval-augmented diffusion models for protein-specific 3D molecule generation. In Ruslan Salakhutdinov, Zico Kolter, Katherine Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp (eds.), *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 20348–20364. PMLR, 21–27 Jul 2024. URL https://proceedings.mlr.press/v235/huang24ab.html.

Clemens Isert, Kenneth Atz, and Gisbert Schneider. Structure-based drug design with geometric deep learning. *Current Opinion in Structural Biology*, 79:102548, 2023. ISSN 0959-440X. doi: https://doi.org/10.1016/j.sbi.2023.102548. URL https://www.sciencedirect.com/science/article/pii/S0959440X23000222.

Ajit Jadhav, Rafaela S. Ferreira, Carleen Klumpp, Bryan T. Mott, Christopher P. Austin, James Inglese, Craig J. Thomas, David J. Maloney, Brian K. Shoichet, and Anton Simeonov. Quantitative analyses of aggregation, autofluorescence, and reactivity artifacts in a screen for inhibitors of a thiol protease. *Journal of Medicinal Chemistry*, 53(1):37–51, 01 2010. ISSN 0022-2623. doi: 10.1021/jm901070c. URL https://doi.org/10.1021/jm901070c.

Xiaohong Ji, Zhen Wang, Zhifeng Gao, Hang Zheng, Linfeng Zhang, Guolin Ke, and Weinan E. Exploring molecular pretraining model at scale. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang (eds.), *Advances in Neural Information Processing Systems*, volume 37, pp. 46956–46978. Curran Associates, Inc., 2024. doi: 10.52202/079017-1489. URL https://proceedings.neurips.cc/paper_files/paper/2024/file/53923bb44655a7defb31c7744c01b62b-Paper-Conference.pdf.

Yinjun Jia, Bowen Gao, Jiaxin Tan, Jiqing Zheng, Xin Hong, Wenyu Zhu, Haichuan Tan, Yuan Xiao, Liping Tan, Hongyi Cai, Yanwen Huang, Zhiheng Deng, Xiangwei Wu, Yue Jin, Yafei Yuan, Jiekang Tian, Wei He, Weiying Ma, Yaqin Zhang, Lei Liu, Chuangye Yan, Wei Zhang, and Yanyan Lan. Deep contrastive learning enables genome-wide virtual screening. *Science*, 391 (6781):eads9530, 2026. doi: 10.1126/science.ads9530. URL https://www.science.org/doi/abs/10.1126/science.ads9530.

Nandan Joshi and Erhan Guven. Improved molecular generation through attribute-driven integrative embeddings and gan selectivity. In *2025 IEEE Conference on Artificial Intelligence (CAI)*, pp. 463–468, 2025. doi: 10.1109/CAI64502.2025.00085.

Sunghwan Kim, Jie Chen, Tiejun Cheng, Asta Gindulyte, Jia He, Siqian He, Qingliang Li, Benjamin A Shoemaker, Paul A Thiessen, Bo Yu, Leonid Zaslavsky, Jian Zhang, and Evan E Bolton. Pubchem 2025 update. *Nucleic Acids Research*, 53(D1):D1516–D1525, 01 2025. ISSN 1362-4962. doi: 10.1093/nar/gkae1059. URL https://doi.org/10.1093/nar/gkae1059.

Craig Knox, Mike Wilson, Christen M Klinger, Mark Franklin, Eponine Oler, Alex Wilson, Allison Pon, Jordan Cox, Na Eun (Lucy) Chin, Seth A Strawbridge, Marysol Garcia-Patino, Ray Kruger, Aadhavya Sivakumaran, Selena Sanford, Rahil Doshi, Nitya Khetarpal, Omolola Fatokun, Daphnee Doucet, Ashley Zubkowski, Dorsa Yahya Rayat, Hayley Jackson, Karxena Harford, Afia Anjum, Mahi Zakir, Fei Wang, Siyang Tian, Brian Lee, Jaanus Liigand, Harrison Peters, Ruo Qi (Rachel) Wang, Tue Nguyen, Denise So, Matthew Sharp, Rodolfo da Silva, Cyrella Gabriel, Joshua Scantlebury, Marissa Jasinski, David Ackerman, Timothy Jewison, Tanvir Sajed, Vasuk Gautam, and David S Wishart. Drugbank 6.0: the drugbank knowledgebase for 2024. *Nucleic Acids Research*, 52(D1):D1265–D1275, 2024. ISSN 0305-1048. doi: 10.1093/nar/gkad976. URL https://doi.org/10.1093/nar/gkad976.

Greg Landrum, Paolo Tosco, Brian Kelley, et al. Rdkit: Open-source cheminformatics.

Tianxiao Li, Haoran Liu, Hongyu Guo, Mark Gerstein, and Martin Renqiang Min. Disentangled autoencoding equivariant diffusion model for controlled generation of 3d molecules. 17(1):4824, 2026. ISSN 2041-1723. doi: 10.1038/s41467-026-71441-9. URL https://doi.org/10.1038/s41467-026-71441-9.

Yi-Lun Liao, Alexander J. Hoffman, Sabrina C. Shen, Alexandre Duval, Sam Walton Norwood, and Tess Smidt. Equiformerv3: Scaling efficient, expressive, and general se(3)-equivariant graph attention transformers, 2026. URL https://arxiv.org/abs/2604.09130.

Tiqing Liu, Linda Hwang, Stephen K Burley, Carmen I Nitsche, Christopher Southan, W Patrick Walters, and Michael K Gilson. Bindingdb in 2024: a fair knowledgebase of protein-small molecule binding data. *Nucleic Acids Research*, 53(D1):D1633–D1644, 01 2025. ISSN 1362-4962. doi: 10.1093/nar/gkae1075. URL https://doi.org/10.1093/nar/gkae1075.

Edgar López-López, Pamela I. Hernández-Estrada, Angel N. Toto-Vazquez, Diana V. Ávila Martínez, Ola Spjuth, and José L. Medina-Franco. Benign-by-design chemistry: Reinventing ligand-based drug design at the edge of ai. *Drug Discovery Today*, 31(4):104691, 2026. ISSN 1359-6446. doi: https://doi.org/10.1016/j.drudis.2026.104691. URL https://www.sciencedirect.com/science/article/pii/S1359644626000966.

Andreas Mayr, Günter Klambauer, Thomas Unterthiner, Marvin Steijaert, Jörg K. Wegner, Hugo Ceulemans, Djork-Arné Clevert, and Sepp Hochreiter. Large-scale comparison of machine learning methods for drug target prediction on chembl. *Chemical Science*, 9(24):5441–5451, 06 2018. ISSN 2041-6520. doi: 10.1039/c8sc00148k. URL https://doi.org/10.1039/c8sc00148k.

Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In Marina Meila and Tong Zhang (eds.), *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 8162–8171. PMLR, 18–24 Jul 2021. URL https://proceedings.mlr.press/v139/nichol21a.html.

Noel M. O'Boyle, Michael Banck, Craig A. James, Chris Morley, Tim Vandermeersch, and Geoffrey R. Hutchison. Open babel: An open chemical toolbox. *Journal of Cheminformatics*, 3(1):33, Oct 2011. doi: 10.1186/1758-2946-3-33. URL https://doi.org/10.1186/1758-2946-3-33.

Daniele Pala and David E. Clark. Caught between a rock and a hard place: current challenges in structure-based drug design. *Drug Discovery Today*, 29(9):104106, 2024. ISSN 1359-6446. doi: https://doi.org/10.1016/j.drudis.2024.104106. URL https://www.sciencedirect.com/science/article/pii/S1359644624002319.

Krunoslav Lehman Pavasovic, Jakob Verbeek, Giulio Biroli, and Marc Mezard. Classifier-free guidance: From high-dimensional analysis to generalized guidance forms, 2025. URL https://arxiv.org/abs/2502.07849.

William Peebles and Saining Xie. Scalable diffusion models with transformers. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 4172–4182, 2023. doi: 10.1109/ICCV51070.2023.00387.

Jian Peng, Jun-Lin Yu, Zeng-Bao Yang, Yi-Ting Chen, Si-Qi Wei, Fan-Bo Meng, Yao-Geng Wang, Xiao-Tian Huang, and Guo-Bo Li. Pharmacophore-oriented 3D molecular generation toward efficient feature-customized drug discovery. *Nature Computational Science*, 5(10):898–914, October 2025. ISSN 2662-8457. doi: 10.1038/s43588-025-00850-5. URL https://doi.org/10.1038/s43588-025-00850-5.

Xingang Peng, Ruihan Guo, Fenglin Guo, Ziyi Wang, Jiayu Sun, Jiaqi Guan, Yinjun Jia, Yan Xu, Yanwen Huang, Muhan Zhang, Jian Peng, Xinquan Wang, Chuanhui Han, Zihua Wang, and Jianzhu Ma. Unified modeling of 3d molecular generation via atomic interactions with PocketX-Mol. 189(7):1904–1922.e28, 2026. ISSN 0092-8674. doi: 10.1016/j.cell.2026.01.003. URL https://doi.org/10.1016/j.cell.2026.01.003.

Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. Film: visual reasoning with a general conditioning layer. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence and Thirtieth Innovative Applications of Artificial Intelligence Conference and Eighth AAAI Symposium on Educational Advances in Artificial Intelligence*, AAAI'18/IAAI'18/EAAI'18. AAAI Press, 2018. ISBN 978-1-57735-800-8.

Rui Qin, Zijie Chen, Yurong Li, Meijing Fang, Longji Shen, Yilong Su, Odin Zhang, Qinghan Wang, Qun Su, Jike Wang, Tingjun Hou, and Yu Kang. Revisiting target-aware de novo molecular generation with tarpass: Between rational design and texas sharpshooter. *Advanced Science*,

13(40):e75411, 2026. doi: https://doi.org/10.1002/advs.75411. URL https://advanced.onlinelibrary.wiley.com/doi/abs/10.1002/advs.75411.

David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 05 2010. ISSN 1549-9596. doi: 10.1021/ci100050t. URL https://doi.org/10.1021/ci100050t.

Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 10684–10695, June 2022.

Daniel M. Rosenbaum, Vadim Cherezov, Michael A. Hanson, Søren G. F. Rasmussen, Foon Sun Thian, Tong Sun Kobilka, Hee-Jung Choi, Xiao-Jie Yao, William I. Weis, Raymond C. Stevens, and Brian K. Kobilka. GPCR engineering yields high-resolution structural insights into β_2-adrenergic receptor function. *Science*, 318(5854):1266–1273, 2007. doi: 10.1126/science.1150609. URL https://www.science.org/doi/abs/10.1126/science.1150609.

Chitwan Saharia, Jonathan Ho, William Chan, Tim Salimans, David J. Fleet, and Mohammad Norouzi. Image super-resolution via iterative refinement. *IEEE Trans. Pattern Anal. Mach. Intell.*, 45(4):4713–4726, April 2023. ISSN 0162-8828. doi: 10.1109/TPAMI.2022.3204461. URL https://doi.org/10.1109/TPAMI.2022.3204461.

Jacob Schnell, Aditya Makkar, Gunadi Gani, Aniket Srinivasan Ashok, Darren Lo, Mike Optis, Alexander Wong, and Yuhao Chen. Composite classifier-free guidance for multi-modal conditioning in wind dynamics super-resolution, 2025. URL https://arxiv.org/abs/2512.13729.

Tiago Sousa, João Correia, and Miguel Pereira, Vítorand Rocha. Generative deep learning for targeted compound design. *Journal of Chemical Information and Modeling*, 61(11):5343–5361, 11 2021. ISSN 1549-9596. doi: 10.1021/acs.jcim.0c01496. URL https://doi.org/10.1021/acs.jcim.0c01496.

Qun Su, Qiaolin Gou, Hui Zhang, Jike Wang, Huiyong Sun, Renling Hu, Rui Qin, Huanxiang Liu, Tingjun Hou, and Yu Kang. LaMGen: LLM-based 3D molecular generation for multi-target drug design. *Nature Communications*, 17(1):5091, April 2026. ISSN 2041-1723. doi: 10.1038/s41467-026-71737-w. URL https://doi.org/10.1038/s41467-026-71737-w.

Yan Sun, Lianghong Chen, Zihao Jing, Yan Yi Li, Dongkyu Kim, Jing-Yan Gao, Reza Noroozi, Grace Y. Yi, Conrard Giresse Tetsassi Feugmo, Anna Klinkova, Kyla Sask, Agustinus Kristiadi, Boyu Wang, Elizabeth R. Gillies, Kun Ping Lu, HaoTian Harvey Shi, and Pingzhao Hu. Generative ai for the design of molecules: Advances and challenges. *Journal of Chemical Information and Modeling*, 65(23):12668–12690, 12 2025. ISSN 1549-9596. doi: 10.1021/acs.jcim.5c02234. URL https://doi.org/10.1021/acs.jcim.5c02234.

Benjamin I. Tingle, Khanh G. Tang, Mar Castanon, John J. Gutierrez, Munkhzul Khurelbaatar, Chinzorig Dandarchuluun, Yurii S. Moroz, and John J. Irwin. ZINC-22: A free multi-billion-scale database of tangible compounds for ligand discovery. *Journal of Chemical Information and Modeling*, 63(4):1166–1176, 02 2023. ISSN 1549-9596. doi: 10.1021/acs.jcim.2c01253. URL https://doi.org/10.1021/acs.jcim.2c01253.

Atabey Ünlü, Elif Çevrim, Melih Gökay Yiğit, Ahmet Sarıgün, Hayriye Çelikbilek, Osman Bayram, Deniz Cansen Kahraman, Abdurrahman Olğaç, Ahmet Sureyya Rifaioglu, Erden Banoğlu, and Tunca Doğan. Target-specific de novo design of drug candidate molecules with graph-transformer-based generative adversarial networks. *Nature Machine Intelligence*, 7(9):1524–1540, Sep 2025. ISSN 2522-5839. doi: 10.1038/s42256-025-01082-y. URL https://doi.org/10.1038/s42256-025-01082-y.

Mihaly Varadi, Stephen Anyango, Mandar Deshpande, et al. AlphaFold protein structure database: massively expanding the structural coverage of protein-sequence space with high-accuracy models. *Nucleic Acids Research*, 50(D1):D439–D444, 2022. doi: 10.1093/nar/gkab1061.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need, 2023. URL https://arxiv.org/abs/1706.03762.

Enrico Ventura, Beatrice Achilli, Luca Ambrogioni, and Carlo Lucibello. Emergence of distortions in high-dimensional guided diffusion models, 2026. URL https://arxiv.org/abs/2602.00716.

Zixu Wang, Yangyang Chen, Pengsen Ma, Zhou Yu, Jianmin Wang, Yuansheng Liu, Xiucai Ye, Tetsuya Sakurai, and Xiangxiang Zeng. Image-based generation for molecule design with sketchmol. *Nature Machine Intelligence*, 7(2):244–255, Feb 2025. ISSN 2522-5839. doi: 10.1038/s42256-025-00982-3. URL https://doi.org/10.1038/s42256-025-00982-3.

Thanapat Worakul, Mohammed Azzouzi, Matthew D. Wodrich, and Clémence Corminboeuf. Modular framework for 3d molecular generation in computational chemistry applications. *Journal of the American Chemical Society*, 148(25):25370–25388, 07 2026. ISSN 0002-7863. doi: 10.1021/jacs.5c19960. URL https://doi.org/10.1021/jacs.5c19960.

wwPDB consortium. Protein data bank: the single global archive for 3d macromolecular structure data. *Nucleic Acids Research*, 47(D1):D520–D528, 01 2019. ISSN 0305-1048. doi: 10.1093/nar/gky949. URL https://doi.org/10.1093/nar/gky949.

Weixin Xie, Jianhang Zhang, Qin Xie, Chaojun Gong, Yuhao Ren, Jin Xie, Qi Sun, Youjun Xu, Luhua Lai, and Jianfeng Pei. Accelerating discovery of bioactive ligands with pharmacophore-informed generative models. 16(1):2391, 2025. ISSN 2041-1723. doi: 10.1038/s41467-025-56349-0. URL https://doi.org/10.1038/s41467-025-56349-0.

Barbara Zdrazil, Eloy Felix, Fiona Hunter, Emma J Manners, James Blackshaw, Sybilla Corbett, Marleen de Veij, Harris Ioannidis, David Mendez Lopez, Juan F Mosquera, Maria Paula Magarinos, Nicolas Bosc, Ricardo Arcila, Tevfik Kizilören, Anna Gaulton, A Patrícia Bento, Melissa F Adasme, Peter Monecke, Gregory A Landrum, and Andrew R Leach. The chembl database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*, 52(D1):D1180–D1192, 01 2024. ISSN 0305-1048. doi: 10.1093/nar/gkad1004. URL https://doi.org/10.1093/nar/gkad1004.

Odin Zhang, Xujun Zhang, Haitao Lin, Cheng Tan, Qinghan Wang, Yuanle Mo, Qiantai Feng, Gang Du, Yuntao Yu, Zichang Jin, Ziyi You, Peicong Lin, Yijie Zhang, Yuyang Tao, Shicheng Chen, Jack Xiaoyu Chen, Chenqing Hua, Weibo Zhao, Runze Ma, Yunpeng Xia, Kejun Ying, Jun Li, Yundian Zeng, Lijun Lang, Peichen Pan, Hanqun Cao, Zihao Song, Bo Qiang, Jiaqi Wang, Pengfei Ji, Lei Bai, Jian Zhang, Chang yu Hsieh, Pheng Ann Heng, Siqi Sun, Tingjun Hou, and Shuangjia Zheng. Odesign: A world model for biomolecular interaction design, 2025. URL https://arxiv.org/abs/2510.22304.

Zaixi Zhang, Wan Xiang Shen, Qi Liu, and Marinka Zitnik. Efficient generation of protein pockets with PocketGen. 6(11):1382–1395, 2024. ISSN 2522-5839. doi: 10.1038/s42256-024-00920-9. URL https://doi.org/10.1038/s42256-024-00920-9.

Qichang Zhao, Haochen Zhao, Linyuan Guo, Kai Zheng, Yajie Li, Qiao Ling, Jing Tang, Yaohang Li, and Jianxin Wang. Coldstartcpi: Induced-fit theory-guided dti predictive model with improved generalization performance. *Nature Communications*, 16(1):6436, Jul 2025a. ISSN 2041-1723. doi: 10.1038/s41467-025-61745-7. URL https://doi.org/10.1038/s41467-025-61745-7.

Yanpeng Zhao, Yuting Xing, Yixin Zhang, Yifei Wang, Mengxuan Wan, Duoyun Yi, Chengkun Wu, Shangze Li, Huiyan Xu, Hongyang Zhang, Ziyi Liu, Guowei Zhou, Mengfan Li, Xuanze Wang, Zhengshan Chen, Ruijiang Li, Lianlian Wu, Dongsheng Zhao, Peng Zan, Song He, and Xiaochen Bo. Evidential deep learning-based drug-target interaction prediction. *Nature Communications*, 16(1):6915, Jul 2025b. ISSN 2041-1723. doi: 10.1038/s41467-025-62235-6. URL https://doi.org/10.1038/s41467-025-62235-6.

APPENDIX

A METHOD DETAILS

A.1 CONTINUOUS DIFFUSION FOR COORDINATES

Coordinates are diffused with a continuous Gaussian process (Ho et al., 2020). The forward process follows

$$X_t = \sqrt{\bar{\alpha}_t} X_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I),$$

where $\bar{\alpha}_t$ follows a cosine schedule (Nichol & Dhariwal, 2021):

$$\bar{\alpha}_t = \frac{f(t)}{f(0)}, \quad f(t) = \cos^2\left(\frac{t+s}{1+s}\cdot\frac{\pi}{2}\right),$$

with offset $s = 0.008$ and $t \in [0, 1]$. The reverse process iteratively denoises from noise back to structured coordinates:

$$X_{t-1} = \frac{1}{\sqrt{\alpha_t}}\Big(X_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(X_t, t)\Big) + \sigma_t z, \quad z \sim \mathcal{N}(0, I),$$

with posterior variance $\sigma_t^2 = \beta_t \dfrac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}$, where $\alpha_t = \bar{\alpha}_t/\bar{\alpha}_{t-1}$, $\beta_t = 1-\alpha_t$, and $\epsilon_\theta(X_t, t)$ is a learned noise predictor. Sampling starts from $X_1 \sim \mathcal{N}(0, I)$ and iteratively denoises to X_0.

Before diffusion, coordinates are centered at the origin to remove translational degrees of freedom. The denoising network $\epsilon_\theta(X_t, t)$ is built on GotenNet (Aykent & Xia, 2025), a SE(3)-equivariant GNN backbone, with time step t encoded via sinusoidal positional embeddings (Vaswani et al., 2023). Training minimizes the noise-prediction mean squared error

$$\mathcal{L}_X = \mathbb{E}_{t, X_0, \epsilon}\big\|\epsilon - \epsilon_\theta(X_t, t)\big\|^2.$$

A.2 DISCRETE DIFFUSION FOR ATOM AND BOND TYPES

The forward process follows a categorical transition (Hoogeboom et al., 2021) that introduces uniform noise with probability β_t,

$$q(c_t \mid c_{t-1}) = \mathcal{C}\left(c_t \mid (1-\beta_t), c_{t-1} + \beta_t/K\right),$$

where $\mathcal{C}$ denotes a categorical distribution and K is the number of classes as described in Section 3.3. The forward marginal from c_0 to c_t follows

$$q(c_t \mid c_0) = \mathcal{C}(c_t \mid \bar{\alpha}_t\, c_0 + (1-\bar{\alpha}_t)/K),$$

where $\alpha_t = 1-\beta_t$ and $\bar{\alpha}_t = \prod_{\tau=1}^{t}\alpha_\tau$.

The reverse process is characterized by the categorical posterior in closed form

$$q(c_{t-1} \mid c_t, c_0) = \mathcal{C}\left(c_{t-1} \mid \theta_{\text{post}}(c_t, c_0)\right), \qquad \theta_{\text{post}}(c_t, c_0) = \frac{\tilde{\theta}}{\sum_{k=1}^{K}\tilde{\theta}_k},$$

$$\tilde{\theta} = [\alpha_t c_t + (1-\alpha_t)/K] \odot [\bar{\alpha}_{t-1} c_0 + (1-\bar{\alpha}_{t-1})/K].$$

The denoising network predicts the clean probability distribution $\hat{c}_0 = \mu_\theta(c_t, t)$ via a softmax output, and the reverse process is parametrized as

$$p_\theta(c_{t-1} \mid c_t) = \mathcal{C}\left(c_{t-1} \mid \theta_{\text{post}}(c_t, \hat{c}_0)\right).$$

Training minimizes the KL divergence between the true posterior and the predicted posterior:

$$\mathcal{L}_c = \mathbb{E}_{t, c_0}\,\mathrm{KL}\big[q(c_{t-1} \mid c_t, c_0)\,\|\,p_\theta(c_{t-1} \mid c_t)\big], \quad c \in \{A, B\},$$

yielding $\mathcal{L}_A$ for atom types and $\mathcal{L}_B$ for bond types. The three diffusion channels are jointly optimized under a hybrid loss $\mathcal{L} = \mathcal{L}_X + \mathcal{L}_A + \mathcal{L}_B$.

B EXPERIMENTAL DETAILS

B.1 DATA CURATION

All datasets were processed in a consistent manner. We performed comprehensive molecular data curation to construct the training dataset. Starting from raw SMILES strings in source databases, we first standardized molecular representations to obtain consistent canonical forms, then applied strict drug-likeness criteria to extract high-quality molecules that improved training stability and generalization.

Molecular Standardization. We preprocessed raw SMILES to ensure consistent representations for reliable filtering and deduplication:

- Extracted parent molecules using ChEMBL Structure Pipeline (Bento et al., 2020) with neutralization
- Adjusted protonation states to pH 7.4 using Open Babel (O'Boyle et al., 2011)
- Generated canonical SMILES with stereochemistry preserved

Composition Constraints. We enforced element composition and ring topology requirements to ensure chemically reasonable structures:

- Allowed elements: C, N, O, S, P, H, F, Cl, Br, I
- Number of heavy atoms: 10–50
- Number of rings: ≤ 5 (aromatic ≤ 5, aliphatic ≤ 2)
- No 3- or 4-membered rings (small rings)
- No rings larger than 6-membered (big rings)

Physicochemical Properties. We applied extended drug-likeness criteria to ensure favorable ADME profiles:

- Hydrogen bond donors: ≤ 6
- Hydrogen bond acceptors: ≤ 10
- Rotatable bonds: ≤ 11
- TPSA: 20–140 Å^2
- LogP: -5 to 6
- Structural alerts: ≤ 2 (Bickerton et al., 2012)
- Chiral centers: ≤ 5

Quality Metrics. We filtered by computational assessments of drug-likeness and synthetic accessibility:

- QED (Quantitative Estimate of Drug-likeness): ≥ 0.28
- SA Score (Synthetic Accessibility): < 6

Substructure Filters. We removed molecules containing problematic substructures or functional groups:

- Passed PAINS_A filter (pan-assay interference compounds) (Baell & Holloway, 2010)
- Passed NIH filter (reactive/promiscuous compounds) (Jadhav et al., 2010)
- Passed ZINC filter (biologically implausible structures) (Tingle et al., 2023)

All filtering criteria were computed using RDKit molecular descriptors and filter catalogs. After filtering and deduplication by canonical SMILES, for ChEMBL as an example, we obtained approximately 1.4 M high-quality drug-like molecules from 2.8 M initial entries for training.

B.2 METRICS

Following TarPass, we evaluate CtrlMol from four complementary perspectives: basic performance, molecular property, binding affinity, and target specificity.

BASIC PERFORMANCE

- **Validity:** Validity measures the proportion of generated molecules that can be successfully parsed and sanitized by RDKit. A high validity score indicates that the model can reliably generate chemically interpretable molecules.
- **Diversity (Uniq@1k):** measures the fraction of unique molecules among the first 1,000 generated molecules, computed as the number of distinct canonical SMILES divided by 1,000.

MOLECULAR PROPERTY

To evaluate whether generated molecules match the physicochemical and structural properties of target-specific active compounds, we follow TarPass and compute distributional distances over 15 molecular descriptors.

The 15 descriptors are listed below.

- **Molecular Weight (MolWt):** Molecular weight measures the overall size of a molecule. It is a basic physicochemical descriptor strongly related to oral bioavailability, docking score bias, and drug-likeness.
- **Heavy Atom Count (HeavyA):** Heavy atom count is the number of non-hydrogen atoms in a molecule. It reflects molecular size while avoiding the variability introduced by implicit or explicit hydrogens.
- **cLogP:** cLogP estimates the octanol-water partition coefficient and measures molecular hydrophobicity. It is closely associated with permeability, solubility, and nonspecific binding risk.
- **Topological Polar Surface Area (TPSA):** TPSA measures the surface area contributed by polar atoms, mainly oxygen and nitrogen atoms. It is commonly used to estimate membrane permeability and oral absorption.
- **Number of Hydrogen-Bond Donors (HBD):** This descriptor counts atoms that can donate hydrogen bonds. It reflects the molecule's ability to form directional polar interactions with protein residues.
- **Number of Hydrogen-Bond Acceptors (HBA):** This descriptor counts atoms that can accept hydrogen bonds. It is important for evaluating potential protein-ligand interaction patterns and drug-likeness.
- **Number of Rotatable Bonds (RotBonds):** The number of rotatable bonds measures molecular flexibility. Molecules with too many rotatable bonds may suffer from high conformational entropy penalties during binding.
- **Fraction of sp3 Carbons (Fsp3):** Fsp3 measures the fraction of carbon atoms with sp3 hybridization. It reflects molecular three-dimensionality and saturation. Higher Fsp3 is often associated with more complex and less planar drug-like structures.
- **Heteroatom Count (HeteroA):** Heteroatom count measures the number of atoms other than carbon and hydrogen, such as N, O, S, P, and halogens. It reflects the potential for polar interactions and functional group diversity.
- **Molecular Volume (MolVol):** Molecular volume estimates the three-dimensional steric size occupied by a molecule and is related to pocket occupancy, molecular shape, and overall ligand bulk.
- **Number of Rings (Rings):** This descriptor counts the total number of ring systems in a molecule. Ring structures are important for molecular rigidity, binding shape complementarity, and drug-like scaffold formation.
- **Number of Molecular Graph Fragments (MGFrags):** MGFrags counts the number of disconnected fragments in the molecular graph, where values greater than one may indicate incomplete or disconnected molecular structures.

- **Normalized Spacial Score (nSPS):** nSPS is a normalized measure of molecular spatial complexity that captures three-dimensional structural features such as hybridization, stereochemistry, ring systems, and molecular branching.
- **Number of Torsions (Torsions):** Torsions measure the number of torsional degrees of freedom in a molecule and reflect conformational flexibility relevant to ligand binding and pose stability.
- **BertzCT Molecular Complexity (BertzCT):** BertzCT is a topological complexity index that measures molecular graph complexity. It captures branching, ring complexity, and overall structural intricacy.

For each descriptor, we compute the Wasserstein distance between the descriptor distribution of generated molecules and that of reference active compounds. The final Wasserstein distance is obtained by averaging over all 15 descriptors. A smaller Wasserstein distance indicates that the generated molecules better match the molecular property distribution of known active compounds for the target.

BINDING AFFINITY

- **Vina docking score:** Binding affinity is evaluated using the Vina docking score under the standardized TarPass docking protocol, where each generated molecule is docked to the corresponding target structure and the top-ranked pose is retained, with lower scores indicating more favorable predicted binding.
- **Ligand efficiency:** Ligand efficiency normalizes the Vina docking score by the number of heavy atoms in the molecule, helping distinguish compact and efficient binders from large molecules that may obtain favorable docking scores mainly due to size-related bias.

TARGET SPECIFICITY

Target specificity evaluates whether the generated molecules are specifically biased toward the chemical space of active compounds for the given target. We use ECFP4 Tanimoto similarity, Fréchet ChemNet Distance, and normalized shift metrics.

- **ECFP4 Tanimoto similarity:** ECFP4 Tanimoto similarity measures the structural similarity between generated molecules and target-specific reference actives by averaging, over generated molecules, the maximum ECFP4 fingerprint similarity to the active set, with higher values indicating closer structural proximity to known target actives.
- **Fréchet ChemNet Distance (FCD) distance:** FCD distance compares the generated and target-active molecular distributions in a learned ChemNet embedding space, with lower values indicating that the generated molecules are distributionally closer to the target-specific active compounds.
- **Fréchet ChemNet Distance (FCD) shift:** FCD shift measures the normalized deviation of the generated molecular distribution from both the target-specific active distribution and the random ChEMBL distribution, where values close to 1 indicate that generated molecules remain within the general drug-like and target-relevant chemical region, while larger values suggest stronger distributional deviation.

In this work, "distance" refers to the direct distributional distance between generated molecules and reference active molecules. In contrast, "shift" refers to the normalized distance of the generated distribution from both the reference active distribution and the random ChEMBL distribution. Distance reflects target-level proximity, while shift reflects whether the generated molecules remain in a plausible drug-like chemical region.

B.3 BASELINES

Following TarPass, we compare CtrlMol with 15 representative molecular generation models spanning three major paradigms: non-3D models, 3D in situ models, and optimization-based methods.

NON-3D MODELS

Non-3D models generate molecular strings, molecular graphs, or building-block combinations without explicitly constructing molecules inside the 3D binding pocket.

- **DeepBlock:** DeepBlock is a building-block-based ligand generation model designed for rational ligand generation with toxicity control. It constructs molecules from chemically meaningful reactive building blocks and benefits from chemical pretraining.
- **DRAGONFLY:** DRAGONFLY is a prospective de novo drug design method based on deep interactome learning. It incorporates target-related information through learned interaction representations and generates molecules without explicit 3D in situ construction.
- **SimpleSBDD:** SimpleSBDD is a simplified structure-based drug design model that generates 2D molecular structures conditioned on target-related information. Although it uses target information, it does not perform full 3D atom-by-atom construction inside the binding pocket.
- **TamGen:** TamGen is a target-aware molecule generation model based on a chemical language model. It generates molecular strings conditioned on target information and benefits from large-scale pretraining on PubChem and CrossDocked2020. Since it operates mainly in molecular string space, it is categorized as a non-3D method.

3D IN SITU MODELS

3D in situ models directly generate molecules inside target binding pockets. These methods explicitly use pocket geometry and are expected to better capture spatial complementarity and protein-ligand interactions. However, they usually require high-quality target structures, incur higher computational cost, and may suffer from pose plausibility issues such as steric clashes, unrealistic conformations, or incomplete molecular graphs.

- **DiffSBDD:** DiffSBDD is an equivariant diffusion model for structure-based drug design. It generates 3D molecular structures conditioned on protein binding pockets and is trained on CrossDocked2020.
- **DrugFlow:** DrugFlow is a flow-based molecular generation model that learns multi-domain distributions for de novo drug design. It operates in 3D binding pockets and generates molecules conditioned on structural information.
- **IPDiff:** IPDiff is a 3D diffusion model that incorporates protein-ligand interaction priors into binding-aware molecular generation. It aims to improve interaction fidelity by explicitly modeling interaction-related information.
- **Lingo3DMol:** Lingo3DMol formulates 3D molecular generation in pockets using a language-modeling framework. It generates molecules conditioned on pocket environments and incorporates interaction-related information during generation.
- **MolCraft:** MolCraft is a structure-based drug design model that generates molecules in a continuous parameter space. It is trained on CrossDocked2020 and belongs to the 3D in situ paradigm.
- **PocketFlow:** PocketFlow is a data- and knowledge-driven structure-based molecular generative model. It generates molecules inside protein pockets and incorporates chemical knowledge constraints to improve chemical validity and plausibility.
- **SurfGen:** SurfGen generates molecules by learning from topological surface and geometric structure information. It explicitly uses protein surface and binding geometry to construct 3D molecules in target pockets.
- **TargetDiff:** TargetDiff is a 3D equivariant diffusion model for target-aware molecular generation and affinity prediction. It generates molecules conditioned on target pocket structures through an equivariant denoising process.

OPTIMIZATION-BASED METHODS

Optimization-based methods further guide molecular generation toward desired objectives such as docking score, binding affinity, drug-likeness, or other molecular properties.

- **DrugFlow-PA:** DrugFlow-PA is an optimized variant of DrugFlow using preference alignment or property-oriented optimization. It is designed to steer the DrugFlow generation process toward more favorable molecular properties or binding-related objectives.
- **MolPilot:** MolPilot is an optimization-based structure-based molecular generation method built upon MolCraft. It introduces a modality-specific optimal schedule to guide molecular generation.
- **REINVENT:** REINVENT is a reinforcement-learning-based molecular generation framework. It is a non-3D model that optimizes generated molecules toward predefined reward functions.

FAIR COMPARISON PROTOCOL

For all baselines, we follow the TarPass evaluation protocol to ensure consistent and fair comparison. Whenever available, we use the official model checkpoints, default hyperparameters, and recommended sampling settings. All generated molecules are processed using the same validation, canonicalization, deduplication, docking, and evaluation workflow as CtrlMol.

B.4 IMPLEMENTATION DETAILS

Main implementations. We trained CtrlMol using the AdamW optimizer with learning rate 1e-4 and batch size 64. To leverage our proposed on-the-fly conformer generation, we randomly sampled 10,000 molecules per epoch and trained for $\sim 12,000$ epochs. The diffusion process used 1,000 steps with cosine noise schedule. We adopted ECFP for molecular condition encoder and DrugCLIP for pocket condition encoder. We set dual-axis CFG dropout rates 0.2, slightly higher than standard single-axis CFG, to balance four possible condition combinations. We implemented CtrlMol using PyTorch. Training took $\sim$ 72 hours on a single H100 GPU with more than 40 GB of memory.

Model architecture. The model totally contains 8.6 M trainable parameters. The architecture consists of input projection layers that encode atomic coordinates, atom types, bond types, and timestep information into 64-dimensional embeddings. Conditions are integrated via proposed ACCE, SACM, and IDA-CFG to modulate the embeddings. The modulated features are then processed by an SE(3)-equivariant GotenNet backbone with 1 attention layer, 8 heads, and 64 hidden dimensions. The maximum SE(3) degree is set to 2. Output projection layers decode the backbone features into predicted coordinates, atom type distributions, and bond type distributions.

On-the-fly conformer generation. For each epoch, we used RDKit's ETKDG algorithm to generate a single 3D conformer for each randomly sampled molecule, followed by MMFF94s optimization with maximum 200 iterations.

Generation validation. Generated molecules with any atomic coordinate outside [-16, 16] Å are considered invalid and discarded. Valid molecules are further filtered by MMFF94s energy Z-score within [-3, 3], and chemical filter catalogs including PAINS_A, NIH, and ZINC. MMFF94s optimization uses a maximum of 1,000 iterations with convergence required.

B.5 TARGET SPECIFICITY

We evaluated FCD between every pair of 18 targets and random to verify target specificity. The diagonal elements represent self-comparisons within the same target, where lower FCD values indicate better target-specificity or target-awareness. Their non-random average corresponds to the Wass. Ref in Table 1. The random-random element corresponds to the minimum FCD achievable for a 1,000-molecule set mentioned in Section 4.3. The off-diagonal elements represent cross-target comparisons, where higher FCD values indicate better target-specificity. The targets comprise 5-HT2A, BCL2, β2AR, BRAF, BRD4, BTK, DRD2, HDAC6, JAK2, MEK1, NAMPT, NQO1, PI3Kα, PPARα, PRMT5, RARα, ROCK1, and TYK2.

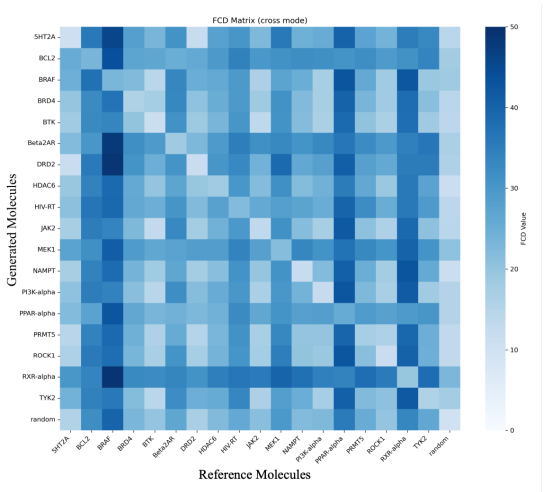


Figure 2: **FCD matrix highlighting target-specific evaluation across 18 targets.** The X-axis and Y-axis denote the reference molecules and the generated molecules across 18 targets and random background, respectively.

B.6 MOLECULAR DESCRIPTORS

We evaluated the property similarity of the generated molecules to the reference molecules using 15 representative molecular descriptors. The average of these 15 descriptors corresponds to Wass. Ref and Wass. Shift in Table 1.

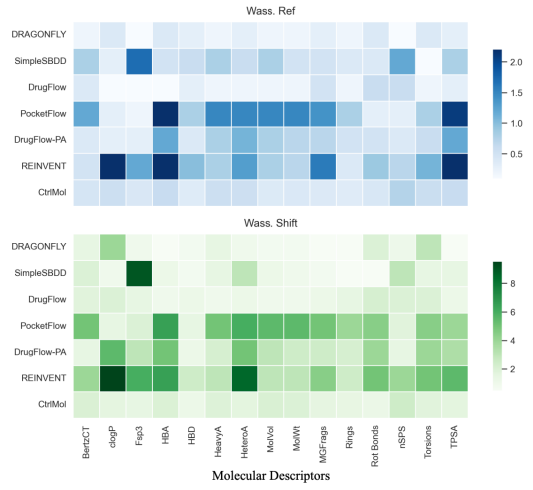


Figure 3: **Wasserstein distances of 15 molecular properties across CtrlMol and baselines.**

B.7 CHEMICAL SPACE

We analyzed the distribution of generated molecules in chemical space across for each of the experiments. Three representative distributions are selected for presentation.

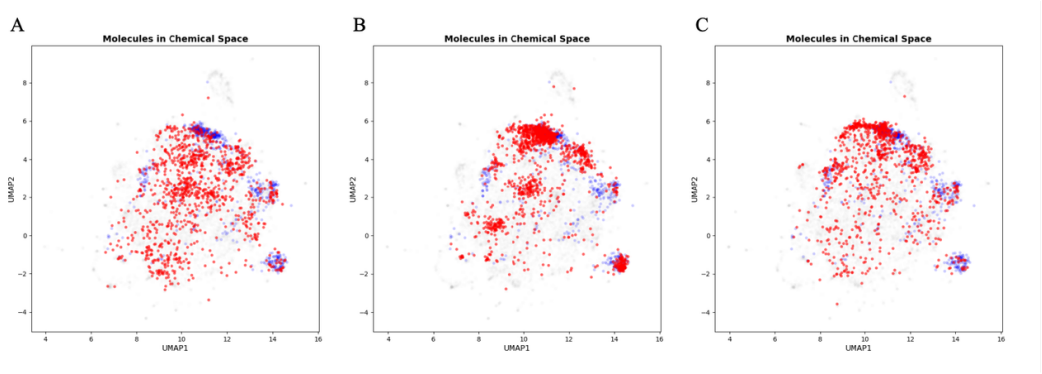


Figure 4: **Generated, reference and random molecules in chemical space.** Generated, reference and random molecules are scattered in red, blue and light gray, respectively. (A) unconditional generated distribution corresponds to $K = 0, L = 0$ in Table 2. (B) overfocused generated distribution corresponds to $K = 1, L = 0$ in Table 2. (C) optimal generated distribution corresponds to $g_{\mathcal{M}} = 2, g_{\mathcal{P}} = 1$ in Table 3.

B.8 DRUG CANDIDATES

We ultimately screened out several ADRB2 drug candidates from the molecules generated by CtrlMol.

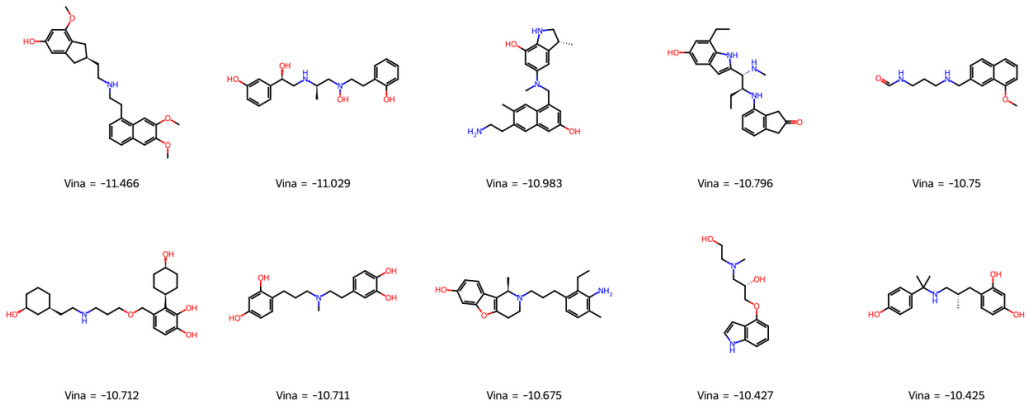


Figure 5: **CtrlMol-designed ADRB2 drug candidates.** The top-10 generated molecules with high binding affinity correspond to $g_{\mathcal{M}} = 2, g_{\mathcal{P}} = 1$ in Table 3.